

Qualification Pack



AI Data Engineering Analyst

QP Code: SSC/Q8106

Version: 4.0

NSQF Level: 5

IT-ITeS SSC NASSCOM || NASSCOM Plot No - 7, 8, 9 & 10, 3rd Floor, Sector 126
Noida Uttar Pradesh - 201303 || email:sscstandards@nasscom.in

Qualification Pack

Contents

SSC/Q8106: AI Data Engineering Analyst	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8154: Understand Database Management and Architecture	5
SSC/N8112: Collect data and conduct data integration to develop a data pipeline	10
SSC/N8119: Ensure data quality and data governance processes and structures are in place	14
DGT/VSQ/N0102: Employability Skills (60 Hours)	18
Assessment Guidelines and Weightage	25
<i>Assessment Guidelines</i>	25
<i>Assessment Weightage</i>	25
Acronyms	27
Glossary	28

Qualification Pack

SSC/Q8106: AI Data Engineering Analyst

Brief Job Description

Data Engineering analyst is responsible for designing, implementing, and maintaining the infrastructure that supports the collection, storage, processing, and analysis of data. They develop and manage data pipelines for extracting, transforming, and loading (ETL) data from various sources while ensuring data quality, accuracy, and security. Key responsibilities include collaborating with stakeholders to identify data needs, managing databases (e.g., MySQL, PostgreSQL), implementing validation and monitoring processes, and utilizing tools like Apache Airflow and Tableau for data management and visualization. Data engineers also enforce data governance and compliance with regulations while continuously improving data processes based on performance metrics and user feedback. Essential skills include proficiency in programming (especially Python and SQL), knowledge of data modeling and database design, and familiarity with cloud platforms and data security practices.

Personal Attributes

A data engineer needs to be well organized and have good attention to detail and problem solving abilities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8154: Understand Database Management and Architecture](#)
2. [SSC/N8112: Collect data and conduct data integration to develop a data pipeline](#)
3. [SSC/N8119: Ensure data quality and data governance processes and structures are in place](#)
4. [DGT/VSQ/N0102: Employability Skills \(60 Hours\)](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
Country	India
NSQF Level	5
Credits	16

Qualification Pack

Aligned to NCO/ISCO/ISIC Code	NCO-2015/2521.0100
Minimum Educational Qualification & Experience	Completed 3 year diploma after 10th (PG/UG/diploma with courses related to Engg./ Science) with 1.5 years of experience in relevant field.Relevant Experience in job roles related to IT/AI/Big Data Analytics/Programming The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications OR Previous relevant Qualification of NSQF Level (4.5) with 1.5 years of experience in relevant field.Relevant Experience in job roles related to IT/AI/Big Data Analytics/Programming The relevant experience would include work, internship, and apprenticeship after completing relevant educational qualifications OR Completed 2nd year of UG (UG Diploma) (PG/UG/diploma with courses related to Engg./ Science)
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	18 Years
Last Reviewed On	NA
Next Review Date	07/10/2028
NSQC Approval Date	07/10/2025
Version	4.0
Reference code on NQR	QG-05-IT-045812025-V2-NASSCOM
NQR Version	2

Qualification Pack

SSC/N8154: Understand Database Management and Architecture

Description

This Nos is about understanding the Database and related architecture

Scope

The scope covers the following :

- Understanding the Database and related architecture

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Collaborate with stakeholders such as data analysts developers and business units to understand data needs usage patterns and performance expectations
- PC2.** Identify specific use cases that the database must support such as reporting analytics or real time data processing and design a logical data model including entities relationships and attributes
- PC3.** Determine the appropriate type of database based on data nature and transform the logical model into physical schema
- PC4.** Install and configure the database management system as per organizational requirements and implement the physical schema
- PC5.** Develop ETL processes to migrate existing data maintaining data integrity and populate the database
- PC6.** Optimize query performance using indexing and query refinement and adjust database settings based on workload needs
- PC7.** Establish role based access controls to secure sensitive data and develop recovery strategies to ensure business continuity
- PC8.** Conduct routine maintenance plan for horizontal or vertical scaling strategies to accommodate future needs
- PC9.** Implement data integrity constraints to ensure data accuracy and perform periodic audits to detect anomalies maintaining data quality and consistency
- PC10.** Document the database architecture including schema diagrams data models and configurations for future reference

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understanding data governance policies procedures and guidelines and knowing how to access information from data sources
- KU2.** Familiarity with organizational policies for sharing and documenting data governance processes and involving the right stakeholders

Qualification Pack

- KU3.** Knowledge of different database types data modeling SQL skills and database management system platforms
- KU4.** Skills in ETL processes performance tuning access control backup and recovery and database monitoring and maintenance
- KU5.** Insights into scalability practices data integrity documentation collaboration and continuous learning for effective database management

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable course of action
- GS2.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Collaborate with stakeholders such as data analysts developers and business units to understand data needs usage patterns and performance expectations	3	5	-	2
PC2. Identify specific use cases that the database must support such as reporting analytics or real time data processing and design a logical data model including entities relationships and attributes	3	5	-	2
PC3. Determine the appropriate type of database based on data nature and transform the logical model into physical schema	3	5	-	2
PC4. Install and configure the database management system as per organizational requirements and implement the physical schema	3	5	-	2
PC5. Develop ETL processes to migrate existing data maintaining data integrity and populate the database	3	5	-	2
PC6. Optimize query performance using indexing and query refinement and adjust database settings based on workload needs	3	5	-	2
PC7. Establish role based access controls to secure sensitive data and develop recovery strategies to ensure business continuity	3	5	-	2
PC8. Conduct routine maintenance plan for horizontal or vertical scaling strategies to accommodate future needs	3	5	-	2
PC9. Implement data integrity constraints to ensure data accuracy and perform periodic audits to detect anomalies maintaining data quality and consistency	3	5	-	2
PC10. Document the database architecture including schema diagrams data models and configurations for future reference	3	5	-	2

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8154
NOS Name	Understand Database Management and Architecture
Sector	IT-ITeS
Sub-Sector	
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	4
Version	1.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8112: Collect data and conduct data integration to develop a data pipeline

Description

This unit is about developing integrations for ensuring data availability for analytical or operational uses.

Scope

The scope covers the following :

- Define requirements
- Develop data warehouses, likes and pipelines

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Identify relevant data sources such as databases APIs and files and assess their quality and format
- PC2.** Create a plan for collecting transforming and loading data into the target platform choosing methods like batch or real-time processing
- PC3.** Use scripts or tools to extract data from sources ensuring efficiency and compliance with source restrictions
- PC4.** Clean and preprocess data to ensure consistency remove duplicates handle missing values and perform necessary transformations
- PC5.** Load data into the target platform while maintaining system performance and validating data accuracy and completeness
- PC6.** Set up monitoring to detect data issues and document workflows for future reference and collaboration
- PC7.** Continuously monitor and improve the data pipeline based on performance and stakeholder feedback
- PC8.** Work with stakeholders to understand data needs and design the pipeline architecture using appropriate tools and technologies
- PC9.** Develop extraction and transformation modules with error handling and optimize load strategies for downstream processes
- PC10.** Automate workflows with orchestration tools and implement monitoring for reliable and scalable data pipeline operations

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand organizational policies for designing data integrations sharing and documenting processes and identify key stakeholders
- KU2.** Familiarity with tools and techniques for developing data integrations using platforms like Hive Storm and Kafka and understanding various data sources and ETL concepts

Qualification Pack

- KU3.** Learn different methods for testing troubleshooting and populating data warehouses and designing data lakes including query mechanisms and data loading strategies
- KU4.** Know management approaches for data lakes including error handling system availability duplication control and performance criteria for both data integration and hardware
- KU5.** Gain skills in querying data writing scripts across various operating systems and understanding system performance for effective data integration and management

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply problem-solving approaches in different situations
- GS2.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Identify relevant data sources such as databases APIs and files and assess their quality and format	3	5	-	2
PC2. Create a plan for collecting transforming and loading data into the target platform choosing methods like batch or real-time processing	3	5	-	2
PC3. Use scripts or tools to extract data from sources ensuring efficiency and compliance with source restrictions	3	5	-	2
PC4. Clean and preprocess data to ensure consistency remove duplicates handle missing values and perform necessary transformations	3	5	-	2
PC5. Load data into the target platform while maintaining system performance and validating data accuracy and completeness	3	5	-	2
PC6. Set up monitoring to detect data issues and document workflows for future reference and collaboration	3	5	-	2
PC7. Continuously monitor and improve the data pipeline based on performance and stakeholder feedback	3	5	-	2
PC8. Work with stakeholders to understand data needs and design the pipeline architecture using appropriate tools and technologies	3	5	-	2
PC9. Develop extraction and transformation modules with error handling and optimize load strategies for downstream processes	3	5	-	2
PC10. Automate workflows with orchestration tools and implement monitoring for reliable and scalable data pipeline operations	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8112
NOS Name	Collect data and conduct data integration to develop a data pipeline
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

SSC/N8119: Ensure data quality and data governance processes and structures are in place

Description

This unit is about defining data utilization and governance processes for the team or organization.

Scope

The scope covers the following :

- Define data assets
- Define data utilization and governance strategies

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** Work with stakeholders to identify data quality criteria like accuracy and completeness and document benchmarks to measure quality
- PC2.** Analyze data to understand its characteristics and create processes to clean duplicates and fix formatting errors
- PC3.** Standardize data formats such as dates and text and set validation rules to ensure data is correct and consistent
- PC4.** Use scripts to automatically check data during ETL processes detect errors and set up alerts for issues
- PC5.** Improve data quality by adding external data sources while ensuring accuracy and standard practices are followed
- PC6.** Investigate the causes of data issues and set up ongoing monitoring with reports and dashboards to track quality trends
- PC7.** Collect feedback from data users about quality issues and update processes and documentation regularly
- PC8.** Create regular data quality reports for stakeholders and stay current with best practices and tools in data quality management
- PC9.** Establish a data governance framework with roles and policies to manage data classification access and responsibilities
- PC10.** Implement security measures like encryption and access controls and review logs to ensure compliance and respond to incidents

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** Understand organizational policies for defining data governance processes sharing and documenting data standards and knowing involved stakeholders
- KU2.** Learn how to identify and access internal and external data assets to meet business requirements

Qualification Pack

- KU3.** Recognize different data assets and establish standards to ensure data quality and integrity
- KU4.** Understand data governance structures and processes and develop standards for effective data utilization
- KU5.** Gain skills in working across various operating systems such as Linux Ubuntu and Windows for managing data assets and governance tasks

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** apply problem-solving approaches in different situations
- GS2.** apply balanced judgments to different situations

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	30	50	-	20
PC1. Work with stakeholders to identify data quality criteria like accuracy and completeness and document benchmarks to measure quality	3	5	-	2
PC2. Analyze data to understand its characteristics and create processes to clean duplicates and fix formatting errors	3	5	-	2
PC3. Standardize data formats such as dates and text and set validation rules to ensure data is correct and consistent	3	5	-	2
PC4. Use scripts to automatically check data during ETL processes detect errors and set up alerts for issues	3	5	-	2
PC5. Improve data quality by adding external data sources while ensuring accuracy and standard practices are followed	3	5	-	2
PC6. Investigate the causes of data issues and set up ongoing monitoring with reports and dashboards to track quality trends	3	5	-	2
PC7. Collect feedback from data users about quality issues and update processes and documentation regularly	3	5	-	2
PC8. Create regular data quality reports for stakeholders and stay current with best practices and tools in data quality management	3	5	-	2
PC9. Establish a data governance framework with roles and policies to manage data classification access and responsibilities	3	5	-	2
PC10. Implement security measures like encryption and access controls and review logs to ensure compliance and respond to incidents	3	5	-	2
NOS Total	30	50	-	20

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8119
NOS Name	Ensure data quality and data governance processes and structures are in place
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	5
Credits	5
Version	3.0
Last Reviewed Date	07/10/2025
Next Review Date	07/10/2028
NSQC Clearance Date	07/10/2025

Qualification Pack

DGT/VSQ/N0102: Employability Skills (60 Hours)

Description

This unit is about employability skills, Constitutional values, becoming a professional in the 21st Century, digital, financial, and legal literacy, diversity and Inclusion, English and communication skills, customer service, entrepreneurship, and apprenticeship, getting ready for jobs and career development.

Scope

The scope covers the following :

- Introduction to Employability Skills
- Constitutional values - Citizenship
- Becoming a Professional in the 21st Century
- Basic English Skills
- Career Development & Goal Setting
- Communication Skills
- Diversity & Inclusion
- Financial and Legal Literacy
- Essential Digital Skills
- Entrepreneurship
- Customer Service
- Getting ready for Apprenticeship & Jobs

Elements and Performance Criteria

Introduction to Employability Skills

To be competent, the user/individual on the job must be able to:

- PC1.** identify employability skills required for jobs in various industries
- PC2.** identify and explore learning and employability portals

Constitutional values – Citizenship

To be competent, the user/individual on the job must be able to:

- PC3.** recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.
- PC4.** follow environmentally sustainable practices

Becoming a Professional in the 21st Century

To be competent, the user/individual on the job must be able to:

- PC5.** recognize the significance of 21st Century Skills for employment
- PC6.** practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life

Basic English Skills

To be competent, the user/individual on the job must be able to:

Qualification Pack

- PC7.** use basic English for everyday conversation in different contexts, in person and over the telephone
- PC8.** read and understand routine information, notes, instructions, mails, letters etc. written in English
- PC9.** write short messages, notes, letters, e-mails etc. in English

Career Development & Goal Setting

To be competent, the user/individual on the job must be able to:

- PC10.** understand the difference between job and career
- PC11.** prepare a career development plan with short- and long-term goals, based on aptitude

Communication Skills

To be competent, the user/individual on the job must be able to:

- PC12.** follow verbal and non-verbal communication etiquette and active listening techniques in various settings
- PC13.** work collaboratively with others in a team

Diversity & Inclusion

To be competent, the user/individual on the job must be able to:

- PC14.** communicate and behave appropriately with all genders and PwD
- PC15.** escalate any issues related to sexual harassment at workplace according to POSH Act

Financial and Legal Literacy

To be competent, the user/individual on the job must be able to:

- PC16.** select financial institutions, products and services as per requirement
- PC17.** carry out offline and online financial transactions, safely and securely
- PC18.** identify common components of salary and compute income, expenses, taxes, investments etc
- PC19.** identify relevant rights and laws and use legal aids to fight against legal exploitation

Essential Digital Skills

To be competent, the user/individual on the job must be able to:

- PC20.** operate digital devices and carry out basic internet operations securely and safely
- PC21.** use e- mail and social media platforms and virtual collaboration tools to work effectively
- PC22.** use basic features of word processor, spreadsheets, and presentations

Entrepreneurship

To be competent, the user/individual on the job must be able to:

- PC23.** identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research
- PC24.** develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion
- PC25.** identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity

Customer Service

To be competent, the user/individual on the job must be able to:

- PC26.** identify different types of customers
- PC27.** identify and respond to customer requests and needs in a professional manner.

Qualification Pack

PC28. follow appropriate hygiene and grooming standards

Getting ready for apprenticeship & Jobs

To be competent, the user/individual on the job must be able to:

PC29. create a professional Curriculum vitae (Résumé)

PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively

PC31. apply to identified job openings using offline /online methods as per requirement

PC32. answer questions politely, with clarity and confidence, during recruitment and selection

PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. need for employability skills and different learning and employability related portals

KU2. various constitutional and personal values

KU3. different environmentally sustainable practices and their importance

KU4. Twenty first (21st) century skills and their importance

KU5. how to use English language for effective verbal (face to face and telephonic) and written communication in formal and informal set up

KU6. importance of career development and setting long- and short-term goals

KU7. about effective communication

KU8. POSH Act

KU9. Gender sensitivity and inclusivity

KU10. different types of financial institutes, products, and services

KU11. how to compute income and expenditure

KU12. importance of maintaining safety and security in offline and online financial transactions

KU13. different legal rights and laws

KU14. different types of digital devices and the procedure to operate them safely and securely

KU15. how to create and operate an e- mail account and use applications such as word processors, spreadsheets etc.

KU16. how to identify business opportunities

KU17. types and needs of customers

KU18. how to apply for a job and prepare for an interview

KU19. apprenticeship scheme and the process of registering on apprenticeship portal

Generic Skills (GS)

User/individual on the job needs to know how to:

GS1. read and write different types of documents/instructions/correspondence

GS2. communicate effectively using appropriate language in formal and informal settings

Qualification Pack

- GS3.** behave politely and appropriately with all
- GS4.** how to work in a virtual mode
- GS5.** perform calculations efficiently
- GS6.** solve problems effectively
- GS7.** pay attention to details
- GS8.** manage time efficiently
- GS9.** maintain hygiene and sanitization to avoid infection

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Introduction to Employability Skills</i>	1	1	-	-
PC1. identify employability skills required for jobs in various industries	-	-	-	-
PC2. identify and explore learning and employability portals	-	-	-	-
<i>Constitutional values – Citizenship</i>	1	1	-	-
PC3. recognize the significance of constitutional values, including civic rights and duties, citizenship, responsibility towards society etc. and personal values and ethics such as honesty, integrity, caring and respecting others, etc.	-	-	-	-
PC4. follow environmentally sustainable practices	-	-	-	-
<i>Becoming a Professional in the 21st Century</i>	2	4	-	-
PC5. recognize the significance of 21st Century Skills for employment	-	-	-	-
PC6. practice the 21st Century Skills such as Self-Awareness, Behaviour Skills, time management, critical and adaptive thinking, problem-solving, creative thinking, social and cultural awareness, emotional awareness, learning to learn for continuous learning etc. in personal and professional life	-	-	-	-
<i>Basic English Skills</i>	2	3	-	-
PC7. use basic English for everyday conversation in different contexts, in person and over the telephone	-	-	-	-
PC8. read and understand routine information, notes, instructions, mails, letters etc. written in English	-	-	-	-
PC9. write short messages, notes, letters, e-mails etc. in English	-	-	-	-
<i>Career Development & Goal Setting</i>	1	2	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
PC10. understand the difference between job and career	-	-	-	-
PC11. prepare a career development plan with short- and long-term goals, based on aptitude	-	-	-	-
<i>Communication Skills</i>	2	2	-	-
PC12. follow verbal and non-verbal communication etiquette and active listening techniques in various settings	-	-	-	-
PC13. work collaboratively with others in a team	-	-	-	-
<i>Diversity & Inclusion</i>	1	2	-	-
PC14. communicate and behave appropriately with all genders and PwD	-	-	-	-
PC15. escalate any issues related to sexual harassment at workplace according to POSH Act	-	-	-	-
<i>Financial and Legal Literacy</i>	2	3	-	-
PC16. select financial institutions, products and services as per requirement	-	-	-	-
PC17. carry out offline and online financial transactions, safely and securely	-	-	-	-
PC18. identify common components of salary and compute income, expenses, taxes, investments etc	-	-	-	-
PC19. identify relevant rights and laws and use legal aids to fight against legal exploitation	-	-	-	-
<i>Essential Digital Skills</i>	3	4	-	-
PC20. operate digital devices and carry out basic internet operations securely and safely	-	-	-	-
PC21. use e- mail and social media platforms and virtual collaboration tools to work effectively	-	-	-	-
PC22. use basic features of word processor, spreadsheets, and presentations	-	-	-	-

Qualification Pack

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Entrepreneurship</i>	2	3	-	-
PC23. identify different types of Entrepreneurship and Enterprises and assess opportunities for potential business through research	-	-	-	-
PC24. develop a business plan and a work model, considering the 4Ps of Marketing Product, Price, Place and Promotion	-	-	-	-
PC25. identify sources of funding, anticipate, and mitigate any financial/ legal hurdles for the potential business opportunity	-	-	-	-
<i>Customer Service</i>	1	2	-	-
PC26. identify different types of customers	-	-	-	-
PC27. identify and respond to customer requests and needs in a professional manner.	-	-	-	-
PC28. follow appropriate hygiene and grooming standards	-	-	-	-
<i>Getting ready for apprenticeship & Jobs</i>	2	3	-	-
PC29. create a professional Curriculum vitae (Résumé)	-	-	-	-
PC30. search for suitable jobs using reliable offline and online sources such as Employment exchange, recruitment agencies, newspapers etc. and job portals, respectively	-	-	-	-
PC31. apply to identified job openings using offline /online methods as per requirement	-	-	-	-
PC32. answer questions politely, with clarity and confidence, during recruitment and selection	-	-	-	-
PC33. identify apprenticeship opportunities and register for it as per guidelines and requirements	-	-	-	-
NOS Total	20	30	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	DGT/VSQ/N0102
NOS Name	Employability Skills (60 Hours)
Sector	Cross Sectoral
Sub-Sector	Professional Skills
Occupation	Employability
NSQF Level	4
Credits	2
Version	1.0
Last Reviewed Date	12/03/2026
Next Review Date	12/03/2029
NSQC Clearance Date	12/03/2026

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

The assessment will consist of a blend of hands-on practical evaluations, viva-voce, and online proctored scenario-based multiple-choice questions ensuring a thorough evaluation of the individual's proficiency in learning outcomes, practical understanding, and real-world application of concepts..

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

Qualification Pack

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8154.Understand Database Management and Architecture	30	50	-	20	100	30
SSC/N8112.Collect data and conduct data integration to develop a data pipeline	30	50	-	20	100	30
SSC/N8119.Ensure data quality and data governance processes and structures are in place	30	50	-	20	100	30
DGT/VSQ/N0102.Employability Skills (60 Hours)	20	30	-	-	50	10
Total	110	180	-	60	350	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.